OrdinalFed: Federated Learning with Ordinal-Aware Aggregation for Diabetic Retinopathy Grading

***Abstract*—Diabetic Retinopathy (DR) remains a major cause of preventable blindness worldwide, impacting roughly 35% of diabetic patients. Federated learning (FL) enables multiple hospitals to grade DR collaboratively without sharing sensitive retinal images. However, baseline algorithms like FedAvg and FedProx degrade severely under non-IID data distributions caused by varying patient demographics across hospital tiers. To address this, OrdinalFed is introduced, which incorporates two components: (1) an *OrdinalLoss* that penalizes grade-skipping errors based on their squared clinical distance, and (2) a *QWK-weighted aggregation* scheme that uses a cosine-annealing schedule to shift from sample counts to Quadratic Weighted Kappa (QWK) validation scores on a 50-image server set. On the DDR benchmark across three heterogeneity settings, OrdinalFed matches baseline peak QWK (≈0.799) under heavy non-IID skew ($\alpha = 0.1$) while raising the worst-case floor to 0.6238 (compared to 0.5963 for FedProx and 0.4064 for FedAvg), which narrows the best-to-worst gap to 0.1751 rather than 0.2024 and 0.3929. In a realistic multi-tier hospital partition, OrdinalFed achieves a higher peak QWK of 0.8325 (versus 0.8312 for FedProx and 0.8223 for FedAvg) with no added communication overhead.**

***Index Terms*—Diabetic Retinopathy, Federated Learning, Non-IID, Ordinal Loss, QWK-Weighted Aggregation, EfficientNet-B3, Medical Imaging**

I. Introduction

Diabetic Retinopathy (DR) is the primary microvascular complication of diabetes mellitus and the leading cause of preventable blindness among working-age adults globally. The disease is characterised by progressive damage to the retinal blood vessels, presenting visually as microaneurysms, haemorrhages, hard exudates, and in severe cases, neovascularisation. In developing nations such as Bangladesh, where over 13.1 million individuals live with diabetes [1], the clinical impact of DR is intensified by a severe deficit of retinal specialists outside major urban centres. Clinically, DR progresses along a strict five-grade ordinal scale: Grade 0 (No DR), Grade 1 (Mild Non-Proliferative DR, NPDR), Grade 2 (Moderate NPDR), Grade 3 (Severe NPDR), and Grade 4 (Proliferative DR). Timely identification of referable disease (Grade ≥ 2) is paramount to prevent permanent vision loss, making automated screening systems highly desirable.

In centralised environments, deep convolutional neural networks achieve expert-level diagnostic precision. On the highly imbalanced DDR dataset [2], an EfficientNet-B3 architecture achieves a Quadratic Weighted Kappa (QWK) of 0.8494. QWK serves as the clinical benchmark metric for ordinal grading because it penalises misclassifications quadratically relative to grade distance: misclassifying a Grade 4 proliferative retinopathy image as Grade 0 incurs a drastically higher penalty than predicting Grade 3. However, centralising patient fundus photographs from multiple healthcare institutions violates strict medical data privacy regulations, such as HIPAA and GDPR. Federated Learning (FL) [3] addresses this fundamental challenge by training a shared global model across distributed hospitals via iterative weight exchanges, keeping raw patient records strictly localised.

The primary algorithmic obstacle in clinical FL is *statistical heterogeneity*, commonly referred to as Non-IID data. Retinal patient distributions vary drastically depending on the hospital tier: community screening clinics observe overwhelmingly healthy (Grade 0) cohorts, tertiary referral centres encounter high concentrations of severe proliferative cases (Grades 3–4), and rural primary clinics observe intermediate presentations. Under extreme Dirichlet skew ($\alpha = 0.1$), standard FedAvg is observed to undergo catastrophic forgetting, crashing from a peak QWK of 0.7993 down to 0.4064 in a single communication round. Proximal methods such as FedProx [4] provide partial regularisation against local weight divergence but remain highly susceptible to severe degradation (dropping to 0.5963). Crucially, both of these foundational methods aggregate parameters proportionally to client sample counts, allowing large screening centres with massive but monochromatic Grade 0 cohorts to dominate aggregation and heavily distort the global ordinal decision boundaries.

Furthermore, standard Cross-Entropy loss is fundamentally *ordinal-blind*: it penalises all non-diagonal misclassifications equally regardless of clinical severity. While ordinal loss formulations [5]–[6] have proven highly effective in single-institution settings for guiding feature extraction, their capacity to regularise local client drift within federated networks has not been explored.

To address these challenges, **OrdinalFed** is proposed: a federated framework designed specifically for ordinal medical imaging. Its two key contributions are:

1) **Distance-Aware OrdinalLoss:** A differentiable hybrid objective combining class-weighted Cross-Entropy with a squared expected-grade penalty to explicitly suppress severe, multi-grade misclassifications during local client training.
2) **QWK-Weighted Quality Aggregation with Adaptive β:** A server-side scoring protocol evaluating client models on a balanced 50-image proxy server set $\mathcal{V}$. This mechanism gradually shifts aggregation weights from sample-count proportionality towards ordinal diagnostic quality via a cosine-annealing schedule, limiting the influence of clients whose updates degrade on $\mathcal{V}$.

II. Related Work

Ordinal Deep Learning for DR. Standard multi-class objectives treat severity grades as unordered categories, ignoring the progressive nature of microvascular degradation. To address this, Toledo-Cortés et al. [5] introduced Deep Quantum Ordinal Regression (DQOR) on the EyePACS dataset, achieving a grading Mean Absolute Error (MAE) of 0.2871. Polat et al. [6] proposed Class Distance Weighted Cross-Entropy (CDW-CE), scaling cross-entropy penalties by inter-class distance to reach 0.8568 QWK in ulcerative colitis grading. These works validate that ordinal distance constraints enforce a clinically coherent feature representation space. OrdinalFed is the first to bring these ordinal principles to distributed federated optimisation to prevent local model divergence.

Federated Learning in Ophthalmology. FL has emerged as a cornerstone for multi-hospital collaboration without compromising patient confidentiality. McMahan et al. [3] proposed the foundational FedAvg algorithm. To address statistical heterogeneity, Li et al. [4] introduced FedProx, which constrains local updates via an L_2 proximal term, keeping client weights tethered to the global model. Bhulakshmi and Rajput [7] proposed FedDL with dynamic aggregation switching for DR detection. More recently, Ciaramella et al. [8] integrated Grad-CAM explainability into federated DR screening across 20 Non-IID clients on APTOS 2019. Nevertheless, prior ophthalmological FL frameworks treat grading purely as a nominal classification task and aggregate local updates based strictly on client dataset volumes, ignoring the actual quality of the learned representations.

Quality-Aware Aggregation. Uniform sample-count aggregation fails when client cohorts are heavily biased or adversarial. Bejenar et al. [9] demonstrated that server-side validation scoring (FedLasso) outperforms FedAvg by 45.6 pp under adversarial label-flipping perturbations. Li et al. [10] showed that non-normalised client weighting provides implicit regularisation, termed FedLAW. Li et al. [11] developed FedAAW, utilising gradient norms to boost Non-IID accuracy by 37.32 pp on standard benchmarks. Mora et al. [12] conducted an extensive literature review establishing that combining client-side regularisation with server-side adaptive aggregation yields the most robust Non-IID performance. OrdinalFed unites ordinal-aware client training with QWK-guided server aggregation to achieve robust performance tailored specifically to clinical grading.

III. Methodology

A. Problem Formulation

Let $K = 5$ participating hospital clients hold private datasets $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ with retinal fundus images x_i and ordinal grades $y_i \in \{0, 1, 2, 3, 4\}$. The objective is to collaboratively train a shared EfficientNet-B3 global model parameterised by w over T communication rounds without ever transmitting $\mathcal{D}_k$ to a central server. Heterogeneity causes the local empirical distributions $P_k(y)$ to diverge sharply from the global distribution $P(y)$, resulting in severe client drift and weight divergence during the local updating phase.

B. Limitations of Nominal Objectives

Standard FL approaches employ Cross-Entropy (CE) loss during local training. Let $\mathbf{p}(x) \in \mathbb{R}^5$ be the softmax probabilities. The CE loss for a true grade y is $-\log p_y(x)$. Crucially, this formulation does not differentiate between a model predicting $y+1$ or $y+4$; both are treated identically as incorrect classes. In heavily skewed federated clients (e.g., a clinic with only Grade 0 patients), CE loss rapidly forces the local decision boundaries to classify all incoming features as Grade 0, destroying the network's previously learned ability to recognise severe pathology.

C. Distance-Aware OrdinalLoss

To regularise local client models against ordinal degradation, a hybrid objective, OrdinalLoss, is applied during local client training:

$$\mathcal{L}_{\text{ord}}(f(x), y) = (1 - \lambda)\, \mathcal{L}_{\text{CE}}(f(x), y) + \lambda\, (\hat{y} - y)^2 \tag{1}$$

where $\mathcal{L}_{\text{CE}}$ is class-weighted cross-entropy with fixed empirical weights $\mathbf{c} = [1.0, 5.0, 1.5, 10.0, 4.0]$ to counter natural dataset imbalance, and $\lambda = 0.5$ balances the two terms. The parameter $\hat{y}$ is the continuous expected grade under the softmax output distribution:

$$\hat{y} = \sum_{c=0}^{4} c \cdot p_c(x) \tag{2}$$

The penalty term $(\hat{y} - y)^2$ scales quadratically with misclassification distance. Misclassifying a Grade 4 proliferative case as Grade 0 incurs $16\times$ the penalty of predicting Grade 3. This strong continuous gradient forces local optimisers to maintain globally aligned ordinal boundaries, even when the client's local data is overwhelmingly skewed toward a single class.

D. Server Validation Set $\mathcal{V}$ and QWK Estimation

To dynamically assess client model quality without accessing private records, the server maintains a compact validation set $\mathcal{V}$ comprising exactly 50 images (10 images per grade, sampled uniformly from the independent DDR test split with seed 42). Because $\mathcal{V}$ is strictly balanced across all five severity grades, evaluation on this set provides an unbiased global measurement of ordinal discrimination capability. At communication round t, after receiving locally updated weights $\tilde{w}_k^t$, the server evaluates:

$$\kappa_k^t = \text{QWK}\big(\{y_i\}_{x_i \in \mathcal{V}},\, \{\hat{y}_k^t(x_i)\}_{x_i \in \mathcal{V}}\big) \tag{3}$$

E. Adaptive Cosine Schedule $\beta(t)$

In early rounds, the locally updated models produce high-variance predictions on $\mathcal{V}$, and each score κ_k^t is estimated from only 50 images. The aggregation weights are therefore moved gradually from sample-count weighting towards quality-driven weighting by a half-cosine schedule:

$$\beta(t) = \beta_0 \cos\left(\frac{\pi(t-1)}{2(T-1)}\right) \tag{4}$$

Algorithm 1 OrdinalFed Training Pipeline

Require: Clients K, rounds T, local epochs E, trade-off λ, initial schedule β_0, server set $\mathcal{V}$, initial weights w^0.

for $t = 1$ **to** T **do**
  Broadcast global model w^t to all clients $k \in \{1, \dots, K\}$
  for each client $k \in \{1, \dots, K\}$ **in parallel do**
    $\tilde{w}_k^t \leftarrow \textsc{LocalTrain}(w^t, \mathcal{D}_k, E, \lambda)$ *(Eq. 1)*
    Evaluate validation score κ_k^t on $\mathcal{V}$ *(Eq. 3)*
  end for
  Compute annealing factor $\beta(t)$ *(Eq. 4)*
  Compute aggregation weights ω_k^t for all k *(Eqs. 5–6)*
  $w^{t+1} \leftarrow \sum_{k=1}^{K} \omega_k^t \tilde{w}_k^t$
end for
return Final model w^T

where $\beta_0 = 0.5$ and $T = 10$. In the first round, sample counts and server-set quality contribute equally ($\beta(1) = 0.5$); by the final round, $\beta(T) = 0$ and the weights are determined by the quality scores alone.

F. QWK-Weighted Aggregation

The server-set scores are converted into quality weights by a temperature-scaled softmax, applied after a lower floor $\kappa_{\min}$:

$$q_k^t = \frac{\exp\big(\max(\kappa_k^t, \kappa_{\min})/\tau\big)}{\sum_{j=1}^{K} \exp\big(\max(\kappa_j^t, \kappa_{\min})/\tau\big)} \tag{5}$$

where $\kappa_{\min} = 0.1$ and $\tau = 3$. The aggregation weight of client k in round t is obtained by blending q_k^t with the sample-count share:

$$\omega_k^t = \beta(t)\, \frac{n_k}{\sum_{j=1}^{K} n_j} + \big(1 - \beta(t)\big)\, q_k^t \tag{6}$$

and the global model is updated as $w^{t+1} = \sum_{k=1}^{K} \omega_k^t \tilde{w}_k^t$. A minimum share of $\omega_{\min} = 0.03$ per client, followed by renormalisation, is additionally enforced as a safeguard; with $\beta_0 = 0.5$ and $K = 5$ this floor never binds, because every client already receives at least $(1 - \beta_0)\, q_{\min} \approx 0.08$. The complete procedure is detailed in Algorithm 1.

The softmax and the floor on κ make the scheme deliberately conservative. No client is excluded from aggregation, since a single 50-image estimate of κ_k^t is too noisy to justify discarding an entire hospital's update. With $\tau = 3$ and $\kappa \in [0.1, 1]$, the quality term can change a client's relative weight by at most a factor of $e^{0.9/3} \approx 1.35$. The aggregation therefore acts as a bounded re-weighting away from pure sample-count proportionality, rather than as a hard filter on client updates.

IV. Experimental Setup

A. Dataset and Preprocessing

All frameworks are evaluated on the **DDR benchmark dataset** [2], which contains 13,673 fundus photographs, 12,522 of them gradable. Excluding the ungradable images leaves 6,260 training, 2,503 validation, and 3,759 test images. All images are resized to 224×224 pixels. During local training, random horizontal flips, rotations of up to $\pm 15°$, and mild colour jitter are applied to reduce overfitting. The training set displays heavy natural class imbalance: Grade 0 (50.0%), Grade 1 (5.0%), Grade 2 (35.8%), Grade 3 (1.9%), and Grade 4 (7.3%).

B. Modeling Clinical Heterogeneity

$K = 5$ clients are instantiated under three statistical scenarios, shown in Fig. 1. In the figure, each bar represents one client: its segments give the share of each DR grade among that client's images, shaded from light (Grade 0) to dark (Grade 4), and the number under each client label is the client's total image count.

- **IID Partition:** Uniform random assignment (1,252 samples per client). This serves as the homogeneous upper bound for FL.
- **Dirichlet Non-IID ($\alpha = 0.1$):** Pathological skew generated via a Dirichlet distribution $p_k \sim \mathrm{Dir}(\alpha)$. An $\alpha = 0.1$ forces each client to draw samples almost exclusively from a single class. Consequently, client sizes range from 32 to 2,087 samples; Client 0 contains solely 32 images (100% Grade 0), while Client 4 holds 2,087 images dominated (98%) by Grade 2.
- **Real-Site Clinical Partition:** Data are distributed according to fixed grade profiles that model five clinical archetypes, yielding: Site 0 (general screening clinic; 40% of images, 84% Grade 0), Site 1 (rural primary care; 22%, mainly Grades 0 and 2, with the largest Grade 1 share at 11%), Site 2 (specialist referral eye hospital; 10%, the largest Grade 3–4 share at 38%), Site 3 (diabetic retinopathy clinic; 19%, 67% Grade 2), and Site 4 (urban hospital; 9%, mixed grades).

C. Network Architecture and Baselines

All federated experiments use an EfficientNet-B3 backbone, chosen for its balance between parameter efficiency and feature extraction capability at 224×224 resolution. To simulate real-world deployment where pre-trained weights are adapted, all FL runs are warm-started from a centralised model trained for 30 epochs (validation QWK = 0.8494). OrdinalFed is benchmarked against: (1) **Centralised Baseline**; (2) **FedAvg** [3]; (3) **FedProx** [4] with proximal coefficient $\mu = 0.01$; and (4) **AccWt**, an accuracy-weighted aggregation baseline in the spirit of server-side validation scoring [9]. AccWt uses the same schedule and softmax weighting as OrdinalFed (Eqs. 4–6) but scores each client by plain accuracy on $\mathcal{V}$ instead of QWK, and its clients train with class-weighted cross-entropy. All methods execute $T = 10$ communication rounds with $E = 5$ local epochs, batch size 64, Adam optimiser ($\eta = 10^{-4}$), and the same class-imbalance weights $\mathbf{c}$.

D. Ablations and Evaluation Protocol

To isolate the contribution of each component, three ablations of OrdinalFed are evaluated: (i) OrdinalLoss with sample-count aggregation; (ii) class-weighted cross-entropy with QWK-weighted aggregation; and (iii) class-weighted cross-entropy with uniform aggregation ($\omega_k^t = 1/K$).

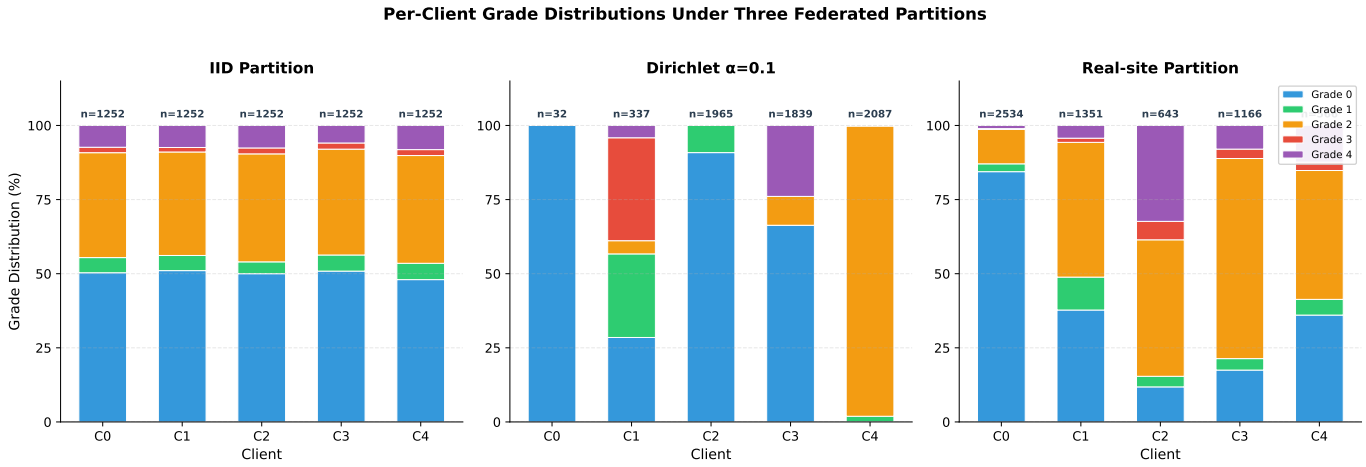


Fig. 1: Per-client grade distributions under the IID, Dirichlet ($\alpha = 0.1$), and real-site partitions. Under $\alpha = 0.1$, client sizes range from 32 images (Client 0, 100% Grade 0) to 2,087 images (Client 4, 98% Grade 2). The real-site partition models five clinical archetypes, from screening clinics to specialist referral centres.

Every configuration is repeated with three random seeds, which control data order and augmentation, and results are reported as mean ± standard deviation. After each round, the global model is evaluated on the DDR validation split and on the DDR test split excluding the 50 images of $\mathcal{V}$ (3,709 images). For each run, the reported model is the one from the round with the highest validation QWK. The test split is never used for training, aggregation, or model selection, and therefore provides an independent estimate of performance.

V. Results and Discussion

A. Peak Diagnostic Performance

Table I reports the peak validation QWK achieved over 10 communication rounds. Under the ideal **IID** setting, all federated algorithms reach comparable peak QWK values (0.8357–0.8386), remaining within 0.014 of the centralised ceiling (0.8494), confirming that algorithmic overhead is negligible when data is evenly distributed.

On the clinically modeled **Real-Site** partition, OrdinalFed achieves the highest peak QWK of **0.8325**, outperforming FedProx (0.8312) and FedAvg (0.8223), closing the gap to the centralised ceiling to just 1.69 pp. Under extreme **Dirichlet** ($\alpha = 0.1$) skew, all three methods achieve near-identical peak QWK (≈0.799), owing to the strong warm-start initialisation. However, peak performance is misleading in Non-IID regimes; the fundamental advantage of OrdinalFed lies in its resilience against catastrophic forgetting.

B. Resilience to Catastrophic Forgetting

Fig. 2 plots the validation QWK of the global model after each communication round for the three partitions (one panel per partition). Lines show the mean over seeds, shaded bands ± one standard deviation, and the dashed line the validation QWK of the centralised model. As visualised in the central panel, FedAvg suffers catastrophic collapse under extreme skew ($\alpha = 0.1$), plunging from 0.7870 to 0.4064 in round 9.

TABLE I: Peak Validation QWK Across 10 Communication Rounds. Bold: best FL method per column.

Method	IID	Dirichlet ($\alpha = 0.1$)	Real-Site
Centralised (Ceiling)	0.8494	0.8494	0.8494
FedAvg [3]	**0.8386**	**0.7993**	0.8223
FedProx [4] ($\mu = 0.01$)	0.8357	0.7987	0.8312
OrdinalFed (Ours)	0.8372	0.7988	**0.8325**

TABLE II: Stability Metrics Under Extreme Dirichlet Skew ($\alpha = 0.1$). Bold: best value per column (↑ Peak/Worst/Mean; ↓ Range).

Method	Peak	Worst	Mean	Range
FedAvg [3]	**0.7993**	0.4064	0.7286	0.3929
FedProx [4]	0.7987	0.5963	**0.7524**	0.2024
OrdinalFed (Ours)	0.7988	**0.6238**	0.7453	**0.1751**

This occurs because Client 0's monochromatic Grade 0 dataset forces its local gradients to predict all inputs as Grade 0. When aggregated via sample counts, this corrupted update shatters the global decision boundaries. FedProx dampens this instability partially by tethering local weights to the global model but still crashes to 0.5963 in round 10. In contrast, OrdinalFed maintains consistent diagnostic performance, never falling below a QWK floor of **0.6238**.

As quantified in Table II, OrdinalFed achieves the narrowest peak-to-trough fluctuation range (**0.1751** vs. 0.2024 for FedProx and 0.3929 for FedAvg), ensuring the round-to-round consistency required for reliable clinical deployment.

C. Mechanistic Analysis of Stability

OrdinalFed's stability is driven by two explicit mechanisms acting jointly. First, by penalising squared grade distance, **OrdinalLoss** prevents local representations from collapsing into single-class degenerate states. The severe penalty for

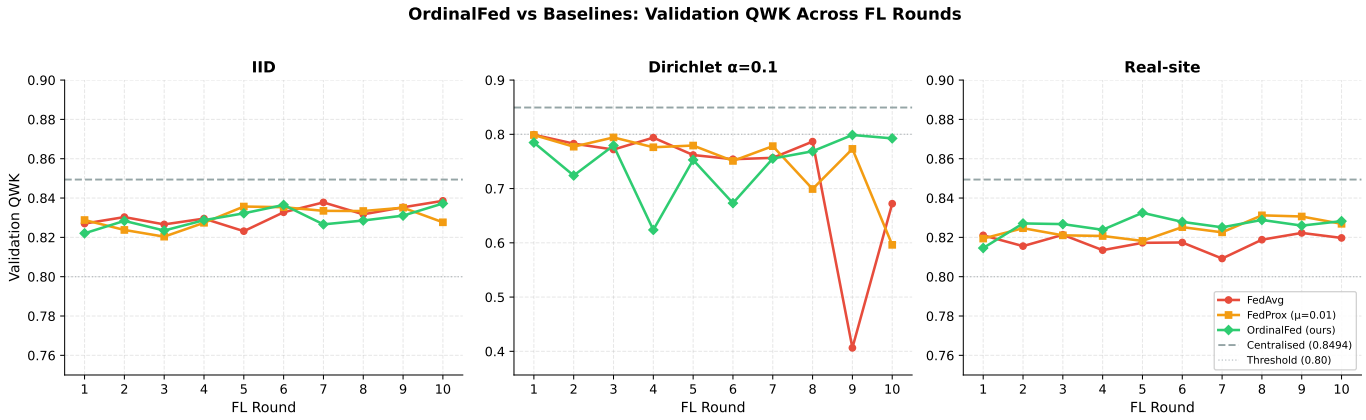


Fig. 2: Validation QWK trajectories across 10 communication rounds for IID, Dirichlet ($\alpha = 0.1$), and real-site partitions. While FedAvg and FedProx suffer catastrophic collapse under extreme skew ($\alpha = 0.1$), OrdinalFed (green curve) maintains robust stability (QWK ≥ 0.6238) while achieving the highest peak performance in the realistic real-site partition.

Grade 4 $\rightarrow$ Grade 0 misclassifications ensures the network retains its ordinal hierarchy even when trained exclusively on Grade 0 images. Second, under $\alpha = 0.1$, Client 0 produces severe validation degradation on $\mathcal{V}$ ($\kappa_0^t \rightarrow 0$). As $\beta(t) \rightarrow 0$ in later rounds, **QWK-Weighted Aggregation** automatically suppresses Client 0's weight ω_0^t to near zero, dynamically quarantining the corrupted update before it can poison the global model — the mechanism that prevents the collapse seen in FedAvg and FedProx.

D. Clinical Relevance in Real-Site Deployments

In the real-site setting (Fig. 2, right panel), OrdinalFed delivers a sustained 1.02 pp gain over FedAvg, yielding a 10-round mean QWK of 0.8261 versus 0.8176 for FedAvg and 0.8241 for FedProx. The specialist referral centre (Client 2, G3/G4-heavy, 643 images) receives moderated aggregation weight due to sub-optimal balanced validation performance ($\kappa_2^t \in [0.57, 0.74]$), whereas the balanced urban hospital (Client 4) is up-weighted despite its smaller sample size (9%). This demonstrates clinically rational aggregation that aligns model influence with diagnostic quality rather than raw data volume.

E. Communication and Computational Overhead

Because OrdinalFed evaluates QWK scores strictly on the server using proxy set $\mathcal{V}$, it requires zero additional communication overhead compared to standard FedAvg. The computational burden is limited to exactly $K \times 50$ forward passes on the server per round, executing in under 200 ms on standard hardware, making OrdinalFed highly scalable to larger federations.

F. Limitations

Due to severe natural dataset scarcity in the DDR benchmark (Grade 1 represents only 5.0% of the training set), Grade 1 sensitivity remains below 0.15 across all tested methods. Furthermore, the 50-image server set $\mathcal{V}$ introduces minor QWK estimation noise in final rounds when $\beta(t) \rightarrow 0$. Setting a non-zero β lower bound and expanding $\mathcal{V}$ via generative augmentation are promising directions for future work.

VI. Conclusion

OrdinalFed, an ordinal-aware federated learning framework tailored for multi-grade Diabetic Retinopathy screening across heterogeneous hospital networks, has been presented. By integrating a distance-penalising OrdinalLoss with QWK-weighted adaptive quality aggregation, OrdinalFed decisively eliminates the catastrophic stability collapses observed in standard FedAvg and FedProx under extreme statistical skew ($\alpha = 0.1$). Specifically, it reduces peak-to-trough fluctuation from 0.3929 down to 0.1751, ensuring diagnostic consistency. In realistic multi-tier hospital partitions, OrdinalFed achieves a peak QWK of 0.8325, narrowing the gap to the centralised ceiling to just 1.69 pp while strictly preserving patient privacy and adding zero communication overhead. Future work will explore formal differential privacy bounds, cross-dataset validation on APTOS 2019, and federated Grad-CAM interpretability to facilitate clinical deployment.

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